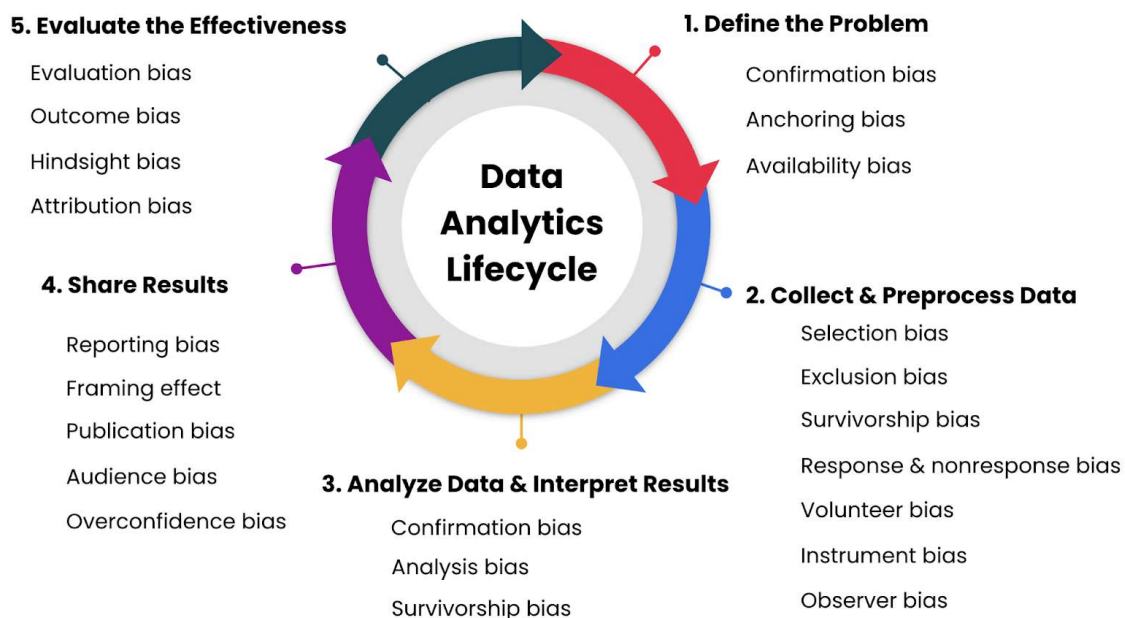


Bias in practice

Bias can unexpectedly creep up into your data and your analysis during any point in the data analytics lifecycle. Bias is a systematic deviation from the truth that can occur during data collection, analysis, interpretation, or review. This systematic nature of bias distinguishes it from random errors, which means it can be **predicted** and often **prevented**, or at the very least, **acknowledged**. Bias can significantly distort findings and lead to incorrect conclusions, impacting the validity of your results.

You have learned about the four most important types of bias in the lecture videos. These are **sampling bias**, **measurement bias**, **response bias**, and **analysis bias**. In practice, there are numerous types of bias, most of which can be broadly categorized into these four main types. The image below illustrates the data analysis lifecycle, a concept you explored in the previous course. Rather than classifying bias solely by type, it can also be categorized based on when it arises within the data analysis lifecycle, as depicted in the image.



How to predict, prevent, and acknowledge bias

By anticipating potential biases and implementing strategies to mitigate them, you can improve the reliability and validity of their results.

Predicting bias

- Understand the common types of biases that can occur at each stage of the life cycle
- Recognize the conditions under which they are most likely to arise.

For example, selection bias is more likely to occur during the data collection stage if the sample is not representative. By being aware of these tendencies, you can proactively design your studies to minimize such risks.

Preventing bias

- Use robust methodologies and best practices throughout the data analytics lifecycle.
 - Employ probabilistic sampling methods to ensure a representative sample
 - Use well-designed data collection instruments to avoid instrument bias
 - Implement blind analysis techniques to reduce observer bias.
- Involve diverse stakeholders to counteract confirmation bias, which is particularly challenging to prevent.
 - Be open to scrutiny
 - Actively seek out opinions that differ from your own
 - Prioritize result validity rather than process or ego
- Regularly refine data preprocessing steps, and scrutinize data and results

Acknowledging bias

Sometimes bias cannot be fully mitigated. Acknowledging the limits of your methods promotes transparency and helps stakeholders understand the coverage of the analysis.

- When presenting results, openly discuss any potential biases that may have influenced the findings and the steps you took to mitigate them.
- Provide a balanced view of the analyses' strengths and limitations to build trust with stakeholders.

Scenario: Analyzing Customer Satisfaction for a Retail Company



Neha, a data analyst at a retail company, has been tasked with improving customer satisfaction. As she leads the project through each stage of the life cycle, Neha considers how to predict, prevent, and acknowledge bias.

Stage 1: Defining the Problem

Neha begins by defining the problem. The company believes that **long checkout times** are the primary reason for **customer dissatisfaction**, but Neha is cautious about **confirmation bias**. She knows that **framing the problem too narrowly** could lead to ineffective solutions. To mitigate this, **Neha involves diverse stakeholders**, including store managers and customers, to gather a comprehensive view of potential dissatisfaction factors. She **frames the problem as an open question**: "What are the primary factors affecting customer satisfaction?"

Potential Biases:

- **Confirmation Bias:** The company believes long checkout times are the main issue, which could lead Neha to initially frame the problem narrowly around this belief, potentially ignoring other factors like product variety or staff behavior.
- **Anchoring Bias:** Initial feedback from a few vocal customers about long checkout times might overly influence Neha's problem definition, causing her to focus too heavily on this aspect.
- **Availability Bias:** Recent complaints about checkout times are more readily available in stakeholders' minds, which might lead Neha to overlook other less obvious but significant issues such as store layout or product availability.

Stage 2: Collecting and Preprocessing Data

Next, Neha **designs a survey** to collect data on customer satisfaction. She is aware that biases can easily occur during this stage and takes steps to **ensure a representative sample**. Neha uses **probabilistic sampling methods** and considers multiple data collection methods (online, in-store, phone) to capture **diverse customer segments**. She also **offers rewards** to encourage participation and **follows up** with non-responders to minimize nonresponse bias.

Potential Biases:

- **Selection Bias:** If Neha only surveys customers who made online purchases, the sample may not represent in-store shoppers, skewing the results.
- **Exclusion Bias:** Even when surveying in the shop, if the survey is done through QR-code on mobile phones, this may on a large scale exclude elderly customers, who are less likely to use online surveys.
- **Convenience Sampling Bias:** Surveying customers who visit the store during weekdays might exclude those who shop on weekends, leading to a non-representative sample.
- **Volunteer Bias:** Customers who volunteer to take the survey might have stronger opinions, either positive or negative, than the average customer, affecting the generalizability of the findings.

Neha also ensures that her **data collection tools are well-designed** to avoid instrument bias. She **trains her team** to minimize observer bias and carefully **frames survey questions** to reduce response bias.

- **Instrument Bias:** Using a poorly designed survey tool that asks leading questions could lead to inaccurate measurements of customer satisfaction.
- **Observer Bias:** If Neha's team members already have set ideas about what the survey should show, their expectations might affect how they understand the responses.
- **Response Bias:** If survey questions are framed in a way that leads participants to provide socially desirable answers, the true picture of customer satisfaction could be distorted.

Stage 3: Analyzing Data and Interpreting Results

When analyzing and interpreting the results, Neha is particularly cautious about confirmation bias and ensures that she **considers all the data**, regardless of whether it supports the initial hypothesis. Neha implements **blind analysis techniques** and involves **multiple analysts** to review the data independently. By doing so, she aims to provide a balanced and objective analysis.

Potential Biases:

- **Confirmation Bias:** Neha must be careful not to focus only on data that supports the belief that checkout times are the main issue, but to evaluate all collected data, regardless of what it shows.
- **Analysis Bias:** If Neha skews the analysis to fit the desired outcome that checkout times are the main issue, she might misinterpret the data, leading to incorrect conclusions.

Stage 4: Sharing Results

Neha knows that **how she presents results can influence perceptions and decisions**. To avoid reporting bias, Neha provides a **balanced view** that includes both positive and negative findings. She is **transparent about any potential biases** that may have influenced the results and discusses the **steps taken to mitigate them**.

Potential Biases:

- **Reporting Bias:** If Neha selectively reports only the positive findings, such as customers appreciating the friendly staff, while leaving out negative feedback about product availability, stakeholders might get a skewed view of customer satisfaction.
- **Framing Effect:** Presenting the results by emphasizing checkout times as the main issue might lead stakeholders to focus only on this aspect, ignoring other important factors.
- **Publication Bias:** If Neha only wants to share results that show significant findings, like a dramatic improvement in satisfaction scores, she might overlook less spectacular but important data.
- **Overconfidence Bias:** If Neha overstates the certainty of her findings, claiming that the survey results definitively pinpoint the issues, stakeholders might over-rely on these results without considering their limitations.

Stage 5: Evaluating Effectiveness

Neha defines clear, **objective criteria for success** and conducts **follow-up analyses** to assess long-term impacts. Neha is mindful of evaluation bias and ensures that she looks for **evidence of both success and failure**. She remains open to revisiting earlier stages if the desired outcomes are not achieved.

Potential Biases:

- **Evaluation Bias:** If Neha looks only for evidence that the changes, such as reducing checkout times, were successful and ignores any negative feedback, she might inaccurately assess the impact.
- **Outcome Bias:** Judging the quality of the decision only based on its outcome, such as an increase in satisfaction scores, rather than the soundness of the decision-making process, could lead to misleading evaluations.
- **Hindsight Bias:** Believing that the decision to focus on checkout times was obviously the right choice after seeing positive results might distort Neha's understanding of the decision-making process.
- **Attribution Bias:** If Neha attributes the success of increased customer satisfaction solely to reduced checkout times, she might overlook other influential factors like improved staff training or product variety.